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Module 4 – Lecture 14

Brownian motion, Poisson processes and Markov dynamics

Today: two noises and their Markov dynamics

Three 50-minute blocks

14:30–15:20: Brownian motion, Wiener space and martingales.
15:35–16:25: Poisson processes, Markov property and transitions.
16:40–17:30: generators, Kolmogorov equations and practice.

- Use independent increments to compute conditional expectations.
- Read matrix evolution of probabilities and expected payoffs.
- Keep the distinction: martingale describes conditional means; Markov describes conditional laws.

Brownian motion: definition

Standard Brownian motion

A real-valued process $(B_t)_{t \geq 0}$ satisfies:

1. $B_0 = 0$ almost surely;
2. its sample paths are continuous almost surely;
3. it has independent increments;
4. $B_t - B_s \sim N(0, t-s)$ for $0 \leq s < t$.

Use its natural filtration $\mathcal{F}_t^B = \sigma(B_u : 0 \leq u \leq t)$. Then $B_t - B_s$ is independent of $\mathcal{F}_s^B$.

Brownian increments and covariance

For $s \le t$, write $B_t = B_s + (B_t - B_s)$. Then

$$\mathbb{E}[B_t] = 0, \quad \mathrm{Var}(B_t) = t, \quad \mathrm{Cov}(B_s, B_t) = s.$$

The covariance identity follows from independence of B_s and the future increment.

Quick computations

$B_5 - B_2 \sim N(0, 3)$, and

$$\mathbb{P}(B_2 - B_1 > 0 \mid \mathcal{F}_1^B) = \frac{1}{2}.$$

B is not stationary, but it has stationary increments.

Finite-dimensional laws of Brownian motion

For $0 \leq t_1 < \cdots < t_m$,

$$(B_{t_1}, \ldots, B_{t_m})$$

is a centered Gaussian vector with covariance matrix

$$\Sigma_{ij} = \min(t_i, t_j).$$

Why this determines the joint law

Every Gaussian vector is determined by its mean and covariance. Brownian dependence across observation times is therefore explicit.

Two times

$\mathrm{Cov}(B_1, B_2) = 1$: Brownian values are dependent even though disjoint increments are independent.

Wiener space: probability on continuous paths

On a fixed horizon $[0, T]$, take

$$\Omega = C_0([0, T]) = \{\omega \in C([0, T]) : \omega(0) = 0\}.$$

Equip it with the Borel sigma-algebra for the uniform norm and Wiener measure W.

- The coordinate process is $B_t(\omega) = \omega(t)$.
- Under W, this coordinate process is Brownian motion.
- One outcome is an entire continuous path, not one Gaussian number.

This is the process-law viewpoint from Lecture 13 in a concrete space.

Brownian martingales: reuse the random-walk proof

For $s < t$, let $Z = B_t - B_s$. It is independent of $\mathcal{F}_s^B$, with $\mathbb{E}[Z] = 0$ and $\mathbb{E}[Z^2] = t - s$. Thus

$$\mathbb{E}[B_t \mid \mathcal{F}_s^B] = B_s,$$

$$\mathbb{E}[B_t^2 \mid \mathcal{F}_s^B] = B_s^2 + t - s.$$

Two basic martingales

B_t and $B_t^2 - t$ are martingales. Both are adapted and integrable; $\mathbb{E}[B_t^2] = t < \infty$.

The correction t has the same role as n in $S_n^2 - n$.

Brownian paths and a stopping-time example

- Brownian paths are continuous but almost surely nowhere differentiable.
- Their total variation is almost surely infinite on every nontrivial bounded interval.
- Classical pathwise differentiation cannot be used for dB_t.

First hitting time

For $a > 0$, $\tau_a = \inf\{t \geq 0 : B_t \geq a\}$ is a stopping time:

$$\{\tau_a \leq t\} = \left\{ \sup_{0 \leq u \leq t} B_u \geq a \right\} \in \mathcal{F}_t^B.$$

Continuity makes the supremum measurable from countably many observed times.

Quadratic variation and Itô calculus are the next module.

Brownian checkpoint

Work for 5 minutes

For $0 < s < t$, compute

1. $\mathbb{E}[B_t - B_s \mid \mathcal{F}_s^B]$;
2. $\mathbb{E}[B_t^2 - t \mid \mathcal{F}_s^B]$;
3. $\mathrm{Var}(B_t - B_s)$;
4. $\mathbb{E}[(B_t - B_s)^2]$.

Explain which step uses independence and which step uses the Gaussian variance.

Brownian checkpoint: answers

$$\mathbb{E}[B_t - B_s \mid \mathcal{F}_s^B] = 0,$$
$$\mathbb{E}[B_t^2 - t \mid \mathcal{F}_s^B] = B_s^2 - s,$$
$$\mathrm{Var}(B_t - B_s) = t - s,$$
$$\mathbb{E}[(B_t - B_s)^2] = t - s.$$

End of Block 1

Independent centered increments give the first martingale; their variance gives the second correction.

Poisson process: counts and jumps

Definition, intensity $\lambda > 0$

A Poisson process starts at $N_0 = 0$, has nondecreasing right-continuous integer-valued paths with unit jumps, has independent increments, and

$$N_t - N_s \sim \text{Poisson}(\lambda(t-s)) \quad (s < t).$$

- λ is the mean number of arrivals per unit of time.
- $\mathbb{E}[N_t] = \text{Var}(N_t) = \lambda t$.
- Use the natural filtration $\mathcal{F}_t^N = \sigma(N_u : 0 \le u \le t)$.

Two count probabilities

For $\lambda = 2$,

$$\mathbb{P}(N_3 = 4) = e^{-6}\frac{6^4}{4!}, \qquad \mathbb{P}(N_5 - N_2 = 0) = e^{-6}.$$

Let $T_1 = \inf\{t : N_t \geq 1\}$. Then

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t}.$$

Hence $T_1 \sim \text{Exp}(\lambda)$ and $\mathbb{E}[T_1] = 1/\lambda$. It is a stopping time because $\{T_1 \leq t\} = \{N_t \geq 1\}$.

Compensated Poisson process

The increment $N_t - N_s$ is independent of $\mathcal{F}_s^N$ and has mean $\lambda(t-s)$. Therefore

$$\mathbb{E}[N_t \mid \mathcal{F}_s^N] = N_s + \lambda(t-s).$$

Subtract the drift

$M_t = N_t - \lambda t$ is adapted, integrable, and

$$\mathbb{E}[M_t \mid \mathcal{F}_s^N] = N_s + \lambda(t-s) - \lambda t = M_s.$$

N_t itself is a submartingale, not a martingale when $\lambda > 0$.

Brownian and Poisson: what changes?

	Brownian B_t	Poisson N_t
Paths	continuous	càdlàg, unit jumps
Increment law	$N(0, h)$	Poisson(λh)
Mean	0	λt
Variance	t	λt
Martingale	B_t	$N_t - \lambda t$
Variation on $[0, T]$	infinite a.s. $(T > 0)$	finite a.s.

Càdlàg means right-continuous with left limits. Both processes have stationary independent increments; neither process is stationary.

The Markov property is about conditional laws

Definition

An adapted process is Markov with respect to $(\mathcal{F}_t)$ if, for every bounded measurable f and $s < t$,

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s].$$

Once the current state is given, the earlier history adds no information about the future law.

Our two examples

$B_t = B_s + (B_t - B_s)$ and $N_t = N_s + (N_t - N_s)$. In each case, the new increment is independent of the past, so the process is Markov in its natural filtration.

Time-homogeneous transitions and Chapman–Kolmogorov

For a time-homogeneous Markov process, write

$$K_t(x, A) = \mathbb{P}(X_{s+t} \in A \mid X_s = x).$$

This depends on elapsed time t, not on starting time s.

Chapman–Kolmogorov

$$K_{s+t}(x, A) = \int_E K_t(y, A)\, K_s(x, dy).$$

Sum or integrate over the intermediate state. For a finite chain with one-step matrix P,

$$p_{ij}^{(2)} = \sum_k p_{ik} p_{kj}, \qquad P^{(n)} = P^n.$$

Finite-state exercise: move a distribution forward

A time-homogeneous chain has

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}, \qquad \pi_0 = (1, 0).$$

Work for 4 minutes

Compute π_1, π_2, and the probability of state 2 after two steps. Explain why row vectors multiply on the left.

Each row of P is a probability vector. The update is $\pi_{n+1} = \pi_n P$.

$$\pi_1 = (0.8, 0.2), \qquad \pi_2 = (0.72, 0.28).$$

The required probability is 0.28.

The same transitions acting on a payoff

For a column vector f of state-dependent payoffs,

$$(Pf)_i = \sum_j p_{ij} f_j = \mathbb{E}[f(X_1) \mid X_0 = i].$$

Probability distributions are row vectors; payoff functions are column vectors. This convention remains in force in continuous time.

For time-homogeneous Markov dynamics, define

$$(P_t f)(x) = \mathbb{E}_x[f(X_t)].$$

Chapman–Kolmogorov gives $P_{s+t} = P_s P_t$ and $P_0 = I$.

Feller: brief introduction

On $E = \mathbb{R}^d$, a Feller semigroup maps $C_0(E)$ into itself and satisfies $\|P_t f - f\|_\infty \to 0$ as $t \downarrow 0$ for each $f \in C_0(E)$.

$C_0(E)$ consists of continuous functions vanishing at infinity. Brownian motion is a standard example. We will not prove Feller theory here.

The generator is an instantaneous expected rate

Definition

For f in the generator domain,

$$Af = \lim_{h \downarrow 0} \frac{P_h f - f}{h}.$$

For a Feller semigroup, this limit is taken in the uniform norm.

The generator describes the initial rate of change of an expected payoff.

Finite-state case

If $P(h) = I + hQ + o(h)$, then $Af = Qf$. All vectors f are in the domain.

Continuous-time chains: rates rather than probabilities

Use one example throughout:

$$Q = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}.$$

- Off-diagonal entries are nonnegative transition rates.
- Rows sum to zero; $q_{ii} = -\sum_{j \neq i} q_{ij} \leq 0$.
- For small h, $\mathbb{P}(X_{t+h} = 2 \mid X_t = 1) = 2h + o(h)$.
- The holding time in state 1 is Exp(2); in state 2 it is Exp(1).

Unlike P, Q is not a matrix of probabilities.

The matrix exponential gives transition probabilities

For a finite-state time-homogeneous chain,

$$P(t) = e^{tQ}, \qquad P(0) = I.$$

For $Q = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$,

$$P(t) = \begin{pmatrix} \frac{1}{3} + \frac{2}{3}e^{-3t} & \frac{2}{3}(1 - e^{-3t}) \\ \frac{1}{3}(1 - e^{-3t}) & \frac{2}{3} + \frac{1}{3}e^{-3t} \end{pmatrix}.$$

- $P(0) = I$; all entries are nonnegative and rows sum to one.
- $P(t + s) = P(t)P(s)$, the continuous-time Chapman–Kolmogorov identity.

We use the formula without deriving the matrix exponential.

Forward Kolmogorov equation: probability mass

Finite-state forward equations

$$P'(t) = P(t)Q, \qquad \pi'(t) = \pi(t)Q.$$

Here $\pi(t) = \pi(0)P(t)$ is a row-vector distribution.

For our two-state example,

$$p_1'(t) = -2p_1(t) + p_2(t), \qquad p_2'(t) = 2p_1(t) - p_2(t).$$

Mass conservation

$p_1'(t) + p_2'(t) = 0$: probability is transferred between the two states.

Backward Kolmogorov equation: expected payoffs

Finite-state backward equation

For a payoff column f, let $u(t) = P(t)f$. Then

$$u'(t) = Qu(t), \qquad u(0) = f.$$

Equivalently, $P'(t) = QP(t)$.

For our two-state example,

$$u_1{}' = 2(u_2 - u_1), \qquad u_2{}' = u_1 - u_2.$$

$u_i(t)$ is the expected payoff after elapsed time t when starting in state i. With a fixed terminal date, $v(s) = P(T - s)f$ satisfies $\partial_s v + Qv = 0$.

Why forward and backward look alike for matrices

Because $P(t) = e^{tQ}$, the matrices $P(t)$ and Q commute:

$$P'(t) = QP(t) = P(t)Q.$$

Keep the objects distinct

Forward: $\pi'(t) = \pi(t)Q$ evolves a distribution.

Backward: $u'(t) = Qu(t)$ evolves an expected payoff as a function of the starting state.

For general state spaces, the forward equation uses the adjoint of the generator; we do not need that theory to solve the finite-state example.

Guided exercise: one generator, two equations

Work for 7 minutes

Let $Q = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$, $\pi(0) = (1, 0)$ and $f = (0, 1)^{\mathsf{T}}$.

1. Check that Q is a valid generator.
2. Write the forward system and solve for $p_2(t)$.
3. Write the backward system for $u(t) = P(t)f$.
4. What is $u_1(t)$? Explain why it equals $p_2(t)$ here.

Solution: forward probabilities and backward payoffs

Off-diagonal entries are nonnegative and rows sum to zero. Since $p_1 = 1 - p_2$,

$$p_2{}' = 2 - 3p_2, \quad p_2(0) = 0, \qquad p_2(t) = \frac{2}{3}(1 - e^{-3t}).$$

The backward system is

$$u_1{}' = 2(u_2 - u_1), \quad u_2{}' = u_1 - u_2, \quad u(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Same number, two viewpoints

$u_1(t) = \mathbb{E}_1[\mathbf{1}_{\{X_t=2\}}] = \mathbb{P}_1(X_t = 2) = p_2(t)$ for the chosen initial state.

Brownian and Poisson generators: a short connection

For smooth compactly supported f, Brownian motion started at x has

$$Af(x) = \frac{1}{2} f''(x).$$

The first-order Taylor term averages to zero; the variance term survives.

For a Poisson process with intensity λ, on finitely supported functions,

$$Af(n) = \lambda(f(n+1) - f(n)).$$

Brownian backward equation

For a smooth bounded payoff, $u(t,x) = \mathbb{E}[f(x + B_t)]$ solves

$$\partial_t u = \tfrac{1}{2} \partial_{xx} u, \qquad u(0,x) = f(x).$$

Core recap and bridge to stochastic calculus

- B_t, $B_t^2 - t$ and $N_t - \lambda t$ are basic martingales in their natural filtrations.
- Markov: the current state determines the future conditional law.
- Chapman–Kolmogorov: compose transitions through intermediate states.
- Generator: instantaneous expected change.
- Forward evolves distributions; backward evolves expected payoffs.

Next module

Quadratic variation, stochastic integration and Itô formula. Local martingales and exponential martingales will be introduced when needed.

Optional reference: invariant distributions

An invariant distribution satisfies

$$\pi P = \pi \quad \text{in discrete time,} \qquad \pi Q = 0 \quad \text{in finite-state continuous time.}$$

For $Q = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$,

$$-2\pi_1 + \pi_2 = 0, \quad \pi_1 + \pi_2 = 1, \quad \pi = (1/3, 2/3).$$

Outside the core lecture

Useful additional reading; not needed for the preceding exercises.

Optional reference: the Brownian forward equation

The transition density from x to y is

$$p(t,x,y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right), \qquad t > 0.$$

It satisfies

$$\partial_t p = \frac{1}{2}\partial_{yy} p \quad \text{(forward)}, \qquad \partial_t p = \frac{1}{2}\partial_{xx} p \quad \text{(backward)}.$$

The initial condition is a point mass: $p(t,x,\cdot)\,dy$ converges weakly to δ_x as $t \downarrow 0$.

Outside the core lecture

The PDE derivation and general adjoint formulation are deferred.